database and system information such as the current time. If a malicious agent can tamper with any of the components in the access control mechanism or with any inputs to the decision, then the malicious agent can subvert the access control mechanism. Even if the components and all of the inputs are collocated within a single file, the operating system security mechanisms are still relied upon to protect the integrity of that file. As discussed, only mandatory security mechanisms can rigorously provide such integrity guarantees.

Even with strong integrity guarantees for the policy decision inputs, if an authorised user invokes malicious software, the malicious software could change an object's security attributes or the policy database's rules without the user's knowledge or consent. The access control mechanism requires a trusted path mechanism in the operating system in order to ensure that arbitrary propagation of access cannot occur without explicit authorisation by a user.

If a malicious agent can impersonate the decider component to the enforcer component, or if a malicious agent can impersonate any source of inputs to the decision, then the malicious agent can subvert the mechanism. If any of the components or external decision input sources are not collocated within a single application, then the access control mechanism requires a protected path mechanism. If a malicious agent can bypass the enforcer component, then it may trivially subvert the access control mechanism. Mandatory security mechanisms in the operating system may be used to ensure that all accesses to the protected objects are mediated by the enforcer component.

Cryptography: An analysis of application-space cryptography may be decomposed into an analysis of the invocation of the cryptographic mechanism and an analysis of the cryptographic mechanism itself. As an initial basis for discussion, suppose that the cryptographic mechanism is a hardware token that implements the necessary cryptographic functions correctly and that there is a secure means by which the cryptographic keys are established in